

{CPU-first time - AT}.

Q#1:-

P_1	P_2	P_3	P_4	P_5	
0	5	8	14	18	20

	Completion-Time	TAT	Waiting time	Response-time
P ₁ s	5	5	0	0
P ₂ s	8	6	3	3
P ₃ s	14	10	4	4
P ₄ s	18	12	8	8
P ₅ s	20	12	10	10

Avg s $\frac{45}{5}$ = 9

Avg s $\frac{25}{5}$ = 5

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Non-pre-emptive:

Response time = Waiting time

select / get process ready to running.

Scheduling length

Time taken by short term scheduler to execute all processes.

$\max(\text{completion-time}) - \min(\text{Arrival-time})$.

Throughput s $\frac{5}{20}$ = $\frac{1}{4}$

Ans

Scheduling-length s $20 - 0$ = 20

Ans.

Job (New)

Ready (Ready-state).

Device (going for I/O)

Separate for every device.

STF-Non-preemptive

Process	Arrival-Time	Burst-Time (BT)	Completion-Time
P ₁	0	7	7
P ₂	2	4	14
P ₃	4	1	8
P ₄	5	4	18
P ₅	6	2	10

P_1	P_3	P_5	P_2	P_4	
0	7	8	10	14	18

Waiting-time	Response-time	Turnaround-time
0	0	7
8	8	12
3	3	4
9	9	13
2	2	4
22	22	40

Avg: $22/5 = 4.4$

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Avg: $40/5 = 8$ *Pr*

Throughput: $5/18 = 0.27$

Response-time: $18 - 0 = 18$ *Pr*